Program for Linear Search

```
#include <iostream>
using namespace std;
int main()
{
        int num[10], val, id=-1;
        for (int i=0; i<10; i++)
        {
        cout<< "\n Enter value " << i+1 <<"= ";
        cin>>num[i];
        }
        cout<< "\n Enter a value to be searched: ";
        cin>>val;
    for (int i=0; i<10; i++)
    {
          if (arr[i] == val)
          {    id= i;
               break;
          }
    }
        if(id==-1)
        cout<< "\n Given value is not found in the array..";
        else
        cout<< "\n The value is found at the position" << id+1;
        return 0;
}
```

The above program reads an array and prompts for the values to be searched. It compares each element of the list with the value that has to be searched until all the elements in the array have been traversed and compared.

Strings

A string is defined as a sequence of characters where each character may be a letter, number or a symbol. Each element occupies one byte of memory. Every string is terminated by a null ('\0', ASCII code 0) character which must be appended at the end of the string. In C++, there is no basic data type to represent a string(However, we can use string object to holding strings in C++). Instead, it implements a string as an one-dimensional character array. When declaring a character array, it also has to hold a null character at the end, and so, the size of the character array should be one character longer than the length of the string.

Character Array (String) creation

To create any kind of array, the size (length) of the array must be known in advance, so that the memory locations can be allocated according to the size of the array. Once an array is created, its length is fixed and cannot be changed during run time. This is shown in figure12.2

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